

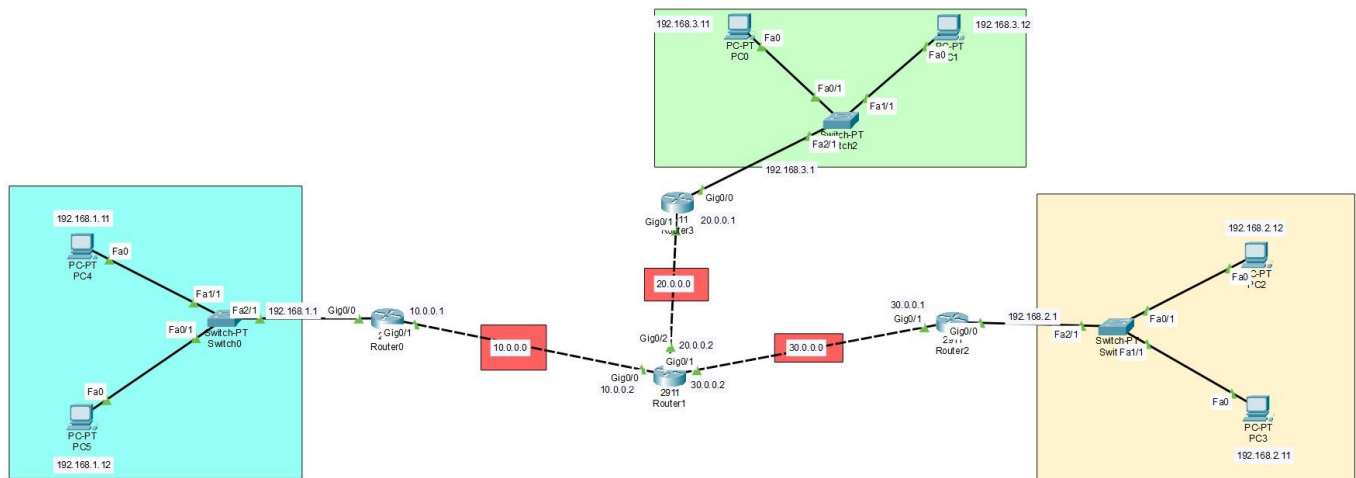
DAY-5

Configuring OSPF Practical :Configure OSPF on a network with multiple areas.
 Task: Simulate a scenario where a router fails and observe OSPF reconvergence.

Q-1 Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

[A]

- To configure **OSPF (Open Shortest Path First)** on a network containing **Area 0 and Area 1**, verify OSPF neighbor relationships, test connectivity between PCs, and observe OSPF behavior during a link failure and recovery.



Device	Interface	IP Address	Network	OSPF Area
R1	G0/0	10.0.0.1/30	10.0.0.0/30	Area 0
R2	G0/0	10.0.0.2/30	10.0.0.0/30	Area 0
R1	G0/1	10.0.1.1/30	10.0.1.0/30	Area 1
R3	G0/0	10.0.1.2/30	10.0.1.0/30	Area 1
R2	G0/1	192.168.10.1/24	192.168.10.0/24	Area 0
R3	G0/1	192.168.20.1/24	192.168.20.0/24	Area 1

PC1	NIC	192.168.10.10/24	192.168.10.0/24	Area 0
PC2	NIC	192.168.20.10/24	192.168.20.0/24	Area 1

Q-2 Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.

[A]

➤ **Router1**

```
enable
configure terminal
```

```
hostname R1
```

```
interface gigabitEthernet 0/0 ip
address 10.0.0.1 255.255.255.252
no shutdown
exit
```

```
interface gigabitEthernet 0/1 ip
address 10.0.1.1 255.255.255.252
no shutdown
exit
```

```
router ospf 1
router-id 1.1.1.1
```

```
network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.3 area 1
```

```
end
write memory
```

➤ **Router2**

```
enable
configure terminal
```

hostname R2

```
interface gigabitEthernet 0/0 ip  
address 10.0.0.2 255.255.255.252  
no shutdown  
exit
```

```
interface gigabitEthernet 0/1 ip  
address 192.168.10.1 255.255.255.0  
no shutdown  
exit
```

```
router ospf 1  
router-id 2.2.2.2
```

```
network 10.0.0.0 0.0.0.3 area 0 network  
192.168.10.0 0.0.0.255 area 0
```

```
end  
write memory
```

➤ **Router3**

```
enable configure  
terminal
```

hostname R3

```
interface gigabitEthernet 0/0 ip  
address 10.0.1.2 255.255.255.252  
no shutdown  
exit
```

```
interface gigabitEthernet 0/1 ip  
address 192.168.20.1 255.255.255.0  
no shutdown  
exit
```

```
router ospf 1  
router-id 3.3.3.3
```

```
network 10.0.1.0 0.0.0.3 area 1
```

```
network 192.168.20.0 0.0.0.255 area 1
```

```
end
```

```
write memory
```

Q-3 Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other.

[A]

➤ **PC1 :**

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

➤ **PC2 :**

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

➤ **Verify the routing table**

• On R2:

```
show ip route
```

You should see an OSPF route toward:
192.168.20.0/24

• On R3:

```
show ip route
```

You should see an OSPF route toward:

192.168.10.0/24

Q-4 Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface. Hint: Track the time it takes for OSPF to update the routing tables after the failure and recovery.

[A]

➤ **Shut down the R1 interface**

enable
configure terminal

interface gigabitEthernet 0/1
shutdown

end

➤ **Check OSPF Neighbor**

- On R1:
show ip ospf neighbor
The R3 neighbor should disappear after OSPF detects the failure.
- On R3:
show ip ospf neighbor
The adjacency with R1 should also disappear.

➤ **Check the Routing Table**

- On R1:

show ip route The route toward:
192.168.20.0/24 will no longer be available
through the failed link.

- On R2:

show ip route

The route toward the Area 1 network will also be removed because there is no longer a path to Area 1.

You can inspect OSPF information with:

show ip ospf and:

show ip ospf interface